



AI can work without internet with this device

A new research project from the University of Central Florida aims to make artificial intelligence (AI) work without the use of the internet. The sophisticated neuromorphic or brain-like devices developed by the researchers are built



UCF NanoScience Technology Center Assistant Professor Tania Roy and Electrical and Computer Engineering doctoral student Adithi Krishnaprasad are working to give technology brain-like computing power (Credit: UCF)

on small, rectangular chips that are about 2.5cm across. Two-dimensional (2D) materials have been used to create transparent, flexible, ultra-thin memristive synapses for neuromorphic computing. By using a different method of production, the researchers were able to change the way the current passes through the device. Their next step will be to improve the technology, which will include the creation of networks with devices that will enable new applications like image recognition.

This robot will help understand how bacteria swims

Brown University students and faculty have constructed a self-propelled robotic swimmer that could aid academics in better understanding the intricate swimming behaviour of bacteria and other microbes. The device is based on the *E. coli* bacterium's shape and was created using a 3D printer. A small engine, a power supply, and other electronics are

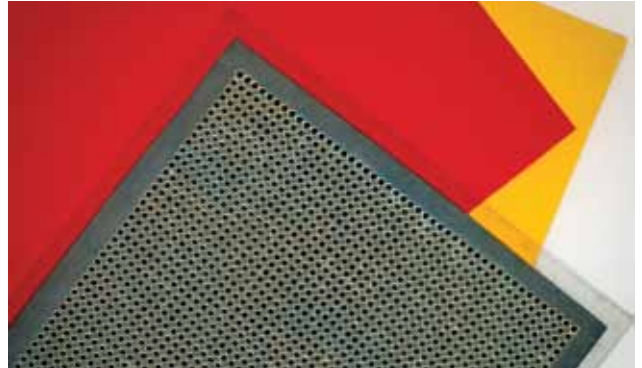


housed in the waterproof head. The engine propels a helical tail which can be changed to experiment with different helix angles and geometries. The motor speed and rotation direction are controlled using a remote control. "What this gives us is the ability to do macroscopic experiments that we have full control over," says Daniel Harris, Assistant Professor of Engineering at Brown.

Self-propelled robotic swimmer (Credit: <https://www.brown.edu>)

A loudspeaker as thin as paper

Engineers at MIT have created a paper-thin loudspeaker that can turn any surface into a sound source. It generates low-distortion sound while consuming a fraction of the energy required by regular loudspeakers. The research simpli-

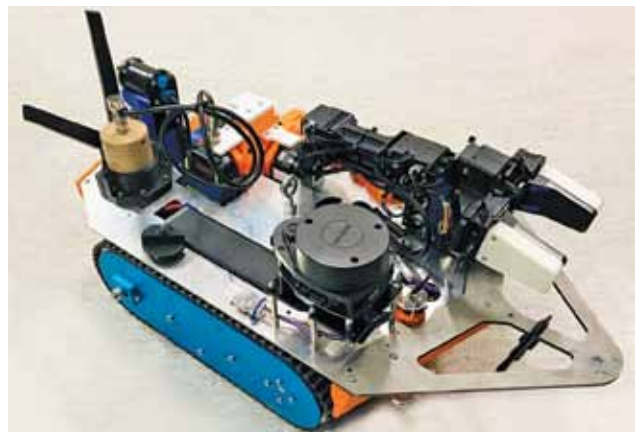


MIT researchers have developed an ultrathin loudspeaker that can turn any rigid surface into a high-quality, active audio source. The straightforward fabrication process they introduced can enable the thin-film devices to be produced at scale (Credit: <https://news.mit.edu>)

fies speaker design by employing a thin layer of shaped piezoelectric material that moves when electricity is applied to it, thus moving the air above it and producing sound. "It feels remarkable to take what looks like a slender sheet of paper, attach two clips to it, plug it into the headphone port of your computer, and start hearing sounds emanating from it," says Vladimir Bulović, the Fariborz Maseeh, lead author of the study.

A robot built for nuclear infrastructure inspection

Lyra, a robot, was employed to investigate a ventilation duct and map radioactive elements in a redundant nuclear laboratory in Dounreay (UK). It was built as a low-cost robot with five radiation detectors, a laser scanner for



Lyra robot for nuclear infrastructure inspection (Credit: www.manchester.ac.uk)